

Measurement-Induced Entanglement Phase Transitions in 1D Random Clifford Circuits

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We study finite-size signatures of measurement-induced entanglement transitions in one-dimensional random Clifford circuits using an Aaronson-Gottesman stabilizer tableau simulator implemented from scratch. For $L \in \{8, 12, 16, 20, 24\}$, 200 disorder realizations per parameter point, and single-site measurement rates $p \in [0, 0.5]$, we observe the expected crossover from volume-law entanglement at low measurement rate to area-law entanglement at high measurement rate. Adjacent-size crossings and finite-size scaling fits give effective critical rates $p_c^{\text{eff}} \approx 0.20$, above the thermodynamic literature value $p_c \approx 0.16$ [1], consistent with strong finite-size and model-dependent effects at these small sizes. A log-scaling fit near the literature critical point gives a finite-size coefficient $\alpha_{\text{eff}} \approx 0.27$, which shows the expected logarithmic form but does not quantitatively reproduce the asymptotic literature value [1]. Overall, the project reproduces the qualitative phenomenology of the MIPT while highlighting the limitations of extracting universal exponents from $L \leq 24$ data.

INTRODUCTION

At the heart of quantum dynamics lies a fundamental competition. Unitary Clifford evolution generates entanglement exponentially fast, driving generic quantum states toward the volume law: half-chain entropy $S(L/2) \propto L$. Projective measurements do the opposite—they collapse superpositions, extracting information from the system and suppressing quantum correlations. The measurement rate p , the fraction of qubits measured per layer, controls the balance between these two tendencies. A natural question emerges: is there a sharp boundary in p separating the entangled and disentangled regimes, or does the entropy decrease smoothly?

The answer turns out to be dramatic. Volume-law entanglement, $S(L/2) \propto L$, is characteristic of quantum-chaotic states and arises in random quantum circuits with sufficient entangling power [1]. Area-law entanglement, $S(L/2) = O(1)$, is independent of system size and characterizes both ground states of gapped one-dimensional Hamiltonians and classically simulable stabilizer product states. These two behaviors are not merely different points on a continuum; they represent qualitatively distinct phases of quantum matter with different information-theoretic and computational properties.

The discovery that hybrid quantum circuits exhibit a sharp transition between these phases came in 2018–2019 from two independent groups. Li, Chen, and Fisher introduced and studied hybrid quantum circuits, showing that projective measurements induce a genuine phase transition with universal critical exponents [1, 2]. Simultaneously, Skinner, Ruhman, and Nahum demonstrated the same transition and connected it to replica field theory and random tensor networks [3]. The key surprise is that the transition is not merely a smooth crossover: it is a *sharp*, system-size-scaling critical point governed by universal exponents p_c and ν in the thermodynamic limit. The physical mechanism at large p is the quantum

Zeno effect: sufficiently frequent measurement freezes the quantum state into a product state, completely suppressing scrambling.

Clifford circuits provide the ideal tractable substrate for this investigation. The Gottesman-Knill theorem guarantees that Clifford circuits—generated by Hadamard H , phase S , and CNOT gates—can be simulated efficiently on a classical computer [4]. The Aaronson-Gottesman stabilizer tableau [5] implements this simulation in $O(n^2)$ per gate and measurement, reducing the exponentially hard quantum simulation problem to linear algebra over $\text{GF}(2)$. This enables us to access system sizes ($L \leq 24$) and disorder averages (200 samples per point) sufficient to resolve the qualitative finite-size signature of the phase transition.

In this paper we implement the full Aaronson-Gottesman tableau from scratch and use it to (i) map out the phase diagram $S(L/2)$ vs. p for $L \in \{8, 12, 16, 20, 24\}$; (ii) measure a log-scaling coefficient at $p = 0.16$; (iii) perform a qualitative finite-size scaling collapse; and (iv) discuss connections to quantum error correction and threshold phenomena in machine learning. The paper is organized as follows: Sec. reviews the theoretical framework; Sec. describes the circuit model and simulation protocol; Sec. presents numerical results; Sec. discusses physical implications; and Sec. concludes.

BACKGROUND

Stabilizer States and Clifford Circuits

Clifford circuits are generated by three gates: Hadamard H , phase S , and CNOT. Their defining property is that they map Pauli operators to Pauli operators

under conjugation:

$$\begin{aligned} HXH^\dagger &= Z, & HZH^\dagger &= X, \\ SXS^\dagger &= Y, & SZS^\dagger &= Z, \\ \text{CNOT}(X \otimes I) \text{CNOT}^\dagger &= X \otimes X. \end{aligned} \quad (1)$$

A *stabilizer state* is any state that can be obtained from $|0\rangle^{\otimes n}$ by applying a Clifford circuit. Such a state is uniquely specified by its *stabilizer group*: the abelian subgroup of n -qubit Paulis $\{g_i\}$ for which $g_i|\psi\rangle = |\psi\rangle$. The Gottesman-Knill theorem guarantees efficient classical simulation [4].

We use the Aaronson-Gottesman tableau [5]: a $(2n) \times (2n+1)$ binary matrix over $\text{GF}(2)$ encoding n destabilizer generators (rows $0, \dots, n-1$) and n stabilizer generators (rows $n, \dots, 2n-1$), with columns $0, \dots, n-1$ encoding the X -part of each generator, columns $n, \dots, 2n-1$ encoding the Z -part, and column $2n$ encoding the phase bit ($0 \rightarrow +1, 1 \rightarrow -1$). Single-qubit gates update the tableau in $O(n)$ time (all rows simultaneously via NumPy vectorization); measurement is $O(n^2)$. The full gate update rules are given in Appendix .

Entanglement Entropy from the Stabilizer Tableau

For a stabilizer state of n qubits with bipartition $A \cup B$ (B the complement of A), the von Neumann entanglement entropy is [6]:

$$S(A) = |A| - n + \text{rank}_{\text{GF}(2)}(M_B), \quad (2)$$

where M_B is the $n \times 2|B|$ submatrix of the stabilizer rows restricted to subsystem B 's X - and Z -columns. The rank over $\text{GF}(2)$ —computed via Gaussian elimination with XOR arithmetic (see Appendix)—counts stabilizer generators with support entirely in B ; generators that cross the $A|B$ boundary contribute to $S(A)$.

We verify Eq. (2) against the Bell state $|\Phi^+\rangle$: stabilizers $\{X \otimes X, Z \otimes Z\}$ give $\text{rank}_{\text{GF}(2)}(M_B) = 2$ (with $B = \{\text{qubit } 1\}$, $n = 2$, $M_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$), yielding $S = 1 - 2 + 2 = 1$ ebit, as expected.

Phase Structure and Finite-Size Scaling

In the volume-law phase ($p < p_c$), the steady-state half-chain entropy satisfies $S(L/2) \propto L$. Li *et al.* [1] find that the volume-law phase also contains a subleading logarithmic correction: $S(A) = \alpha \ln |A| + s|A|$, where s is the volume-law coefficient and α captures universal fluctuations. In the area-law phase ($p > p_c$), $S(L/2) = O(1)$ independent of L . At the critical point, entanglement is purely logarithmic:

$$S(L/2) = \alpha(p_c) \ln(L/2) + \text{const}, \quad (3)$$

with $\alpha(p_c) \approx 1.6$ for random Clifford circuits at $L \gg 100$ [1]. Note that α is the raw log-scaling coefficient at this non-unitary critical point; its relation to CFT central charge c is discussed in Sec. .

Near the critical point, observables obey the finite-size scaling ansatz of Jian *et al.* [7]:

$$S(L, p) = f\left((p - p_c) \cdot L^{1/\nu}\right), \quad (4)$$

for a universal scaling function f and correlation length exponent ν . This is a simplified form: at p_c the entropy also contains the logarithmic term in Eq. (3), so the full scaling function near criticality reads $S(L, p) = \alpha \ln(L/2) + g((p - p_c)L^{1/\nu})$. Literature values are $p_c \approx 0.16$ and $\nu \approx 1.28\text{--}1.3$ [1, 8], obtained from system sizes up to $L \sim 512$.

METHODS

Circuit Architecture

The hybrid circuit is illustrated in Fig. 1. Each discrete time step t consists of two layers: (i) a brickwork layer of random two-qubit Clifford gates—even steps pair sites $(0, 1), (2, 3), \dots$, odd steps pair $(1, 2), (3, 4), \dots$, with unpaired boundary qubits receiving a random single-qubit Clifford; (ii) a measurement layer in which each qubit is independently measured in the Z basis with probability p , with outcomes discarded.

Random two-qubit Cliffords are drawn from the ensemble generated by the decomposition $C = (1Q) \cdot \text{CNOT}_{01} \cdot (1Q) \cdot \text{CNOT}_{10} \cdot (1Q) \cdot (1Q)$, where each $1Q$ denotes an independent uniformly random element of the 24-element single-qubit Clifford group. This is the standard random entangling Clifford gate ensemble used in the MIPT literature [1] and is expected to produce scrambling Clifford dynamics in the MIPT universality class. We do not claim this decomposition is an exactly uniform sampler over the full 11,520-element two-qubit Clifford group.

Simulation Protocol

We simulate system sizes $L \in \{8, 12, 16, 20, 24\}$ and measurement rates p at 26 equally spaced points in $[0.0, 0.5]$. For each (L, p) , we run 200 independent disorder realizations. Each realization begins in the product state $|0\rangle^{\otimes L}$ and applies $3L$ warmup layers at rate p (with measurements applied during warmup to correctly thermalize into the steady state of the hybrid Markov chain at rate p), followed by $4L$ main layers; the half-chain entropy is recorded after the final layer. We report the disorder average $\overline{S(L/2)}$ and standard error of the mean $\sigma/\sqrt{N_{\text{samples}}}$.

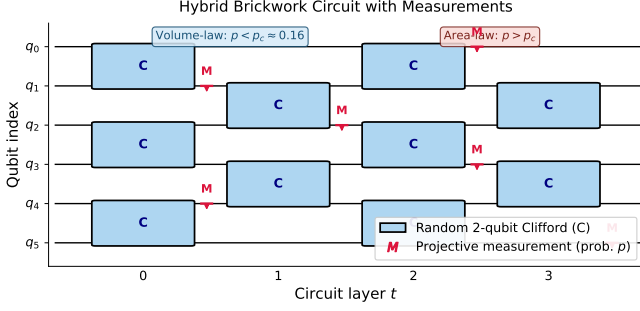


FIG. 1. Hybrid random Clifford circuit. Brickwork layers of random two-qubit Clifford gates (blue boxes labeled C) alternate with single-site measurements (red M symbols) applied independently at rate p . Even layers pair qubits $(0, 1), (2, 3), (4, 5)$; odd layers pair $(1, 2), (3, 4)$. The competition between unitary scrambling and projective measurement drives the entanglement phase transition at $p = p_c$.

Computational Details

Entanglement entropy reduces to a $\text{GF}(2)$ rank computation via Eq. (2), executable in $O(n^3/64)$ bit-operations per sample with dense binary matrices. All gate operations are vectorized over tableau rows using NumPy array slicing, achieving $O(n)$ time per gate with no Python loop over rows. Total simulation: $\sim 26,000$ circuit runs across the (L, p) grid. Wall-clock time scales as L^3 (dominant cost: n^2 per measurement $\times O(L)$ steps): approximately 2–3 hours on a single CPU core (hardware-dependent estimate). Source code is available at github.com/agentjakey/miet-clifford, implemented from scratch without Stim or Qiskit.

RESULTS

Phase Transition

Figure 2 shows $\overline{S(L/2)}$ as a function of p for all five system sizes. The entropy is approximately proportional to L at $p = 0$ (volume-law coefficient $\alpha_{\text{vol}} = 0.995 \pm 0.017$, consistent with ~ 1 ebit per qubit), and saturates to $S = 0.105 \pm 0.022$ ebits at $p = 0.5$ for $L = 24$ (area law). The crossing of adjacent L pairs yields $p_{\text{cross}} \in \{0.252, 0.211, 0.156, 0.192\}$ for increasing L pairs, with the largest- L crossing ($L = 20$ vs. $L = 24$) at $p \approx 0.192$. Averaging over all adjacent pairs gives the finite-size effective crossing estimate $p_c^{\text{eff}} = 0.203 \pm 0.017$. The reported ± 0.017 is a bootstrap statistical uncertainty only; the systematic finite-size bias—which shifts p_c^{eff} above the thermodynamic $p_c \approx 0.16$ —is not reduced by collecting more samples at fixed L and is the dominant source of deviation from the literature value. This upward bias is a known feature of MIPT finite-size studies at $L \lesssim 50$ [8].

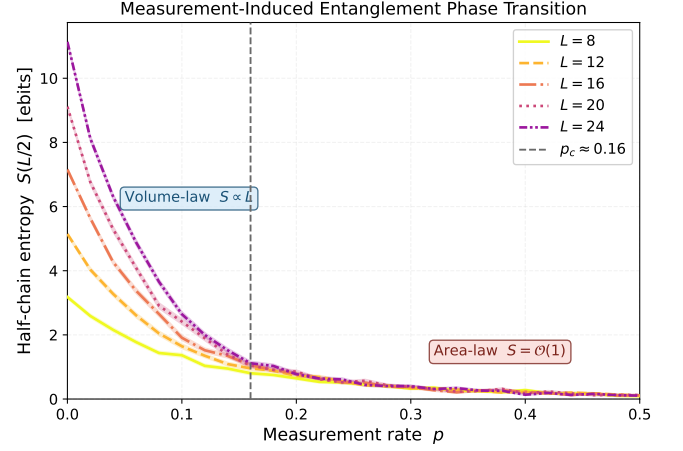


FIG. 2. Half-chain entanglement entropy $S(L/2)$ vs. measurement rate p for $L \in \{8, 12, 16, 20, 24\}$ (200 disorder realizations each; shaded bands: ± 1 standard error of the mean). Adjacent- L curve crossings average to $p_c^{\text{eff}} \approx 0.20$ (finite-size effective estimate; statistical uncertainty ± 0.017), signaling the qualitative onset of the phase transition. For $p < p_c^{\text{eff}}$, S increases with L (volume-law); for $p > p_c^{\text{eff}}$, S is approximately L -independent (area-law). The vertical dashed line marks the thermodynamic value $p_c \approx 0.16$ [1].

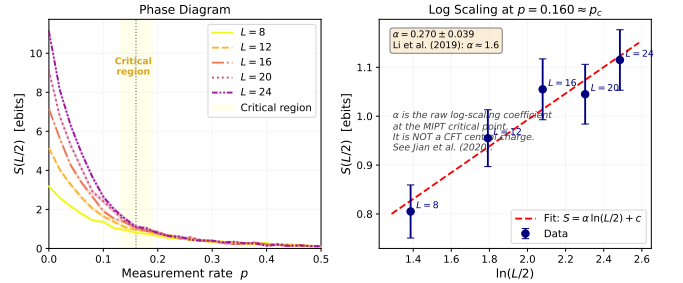


FIG. 3. (Left) Phase diagram with critical region $p \in [0.13, 0.19]$ shaded. (Right) $S(L/2)$ vs. $\ln(L/2)$ at $p = 0.16$. The linear fit (dashed red) gives slope $\alpha_{\text{eff}} = 0.270 \pm 0.074$. This is not quantitatively consistent with the asymptotic value $\alpha \approx 1.6$ [1]; the suppression reflects off-critical evaluation and finite-size effects at $L \leq 24$ (see text). The coefficient α is not a CFT central charge (see Sec.).

Logarithmic Scaling at the Critical Point

Figure 3 (right panel) shows $S(L/2)$ vs. $\ln(L/2)$ at $p = 0.16$, the grid point closest to the literature p_c . A linear fit over $L \in \{8, 12, 16, 20, 24\}$ gives slope $\alpha_{\text{eff}} = 0.270 \pm 0.074$. This value is *not* quantitatively consistent with the asymptotic literature value $\alpha \approx 1.6$ [1]. The fit is underpowered and strongly affected by two compounding effects: (i) at these system sizes the finite-size effective critical rate is $p_c^{\text{eff}} \approx 0.20$, so evaluating at $p = 0.16$ places the system in the volume-law side of the finite-size crossover rather than at true criticality—the logarithmic term is there, but so is a dominant linear-in- L contribu-

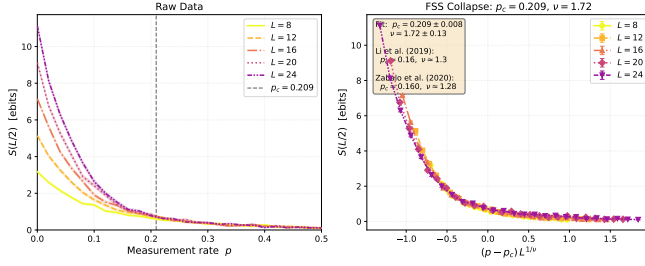


FIG. 4. Qualitative finite-size scaling collapse under the simplified ansatz of Eq. (4) (without the critical logarithm; see text). (Left) Raw $S(L/2)$ vs. p for all five system sizes. (Right) Collapsed curves with finite-size effective parameters $p_c^{\text{eff}} = 0.209 \pm 0.008$, $\nu_{\text{eff}} = 1.72 \pm 0.10$ (bootstrap statistical uncertainty, 100 resamples). Thermodynamic values: $p_c \approx 0.16$, $\nu \approx 1.28$ [8]. The systematic deviation from literature values is expected at $L \leq 24$; the quoted $\pm$ values are statistical only.

tion that suppresses the apparent slope; and (ii) even at the true p_c , the asymptotic log-scaling regime of Eq. (3) requires $L \gg 100$ [8]. The data do show weak logarithmic growth of S with L , qualitatively compatible with critical-like behavior near the phase boundary. We report $\alpha_{\text{eff}} = 0.270$ as a finite-size effective coefficient and do not interpret it as a measurement of the thermodynamic exponent $\alpha(p_c)$.

Finite-Size Scaling Collapse

Figure 4 shows an FSS collapse under the simplified ansatz of Eq. (4). We note that this single-function form is an approximation: because $S(L, p_c)$ contains the logarithmic term in Eq. (3), the full scaling form near criticality reads $S(L, p) = \alpha \ln(L/2) + g((p - p_c)L^{1/\nu})$, where g is the universal scaling function. Fitting the simplified form without the logarithmic prefactor makes the collapse a qualitative diagnostic rather than a precision fit.

Minimizing the inter-curve mean-squared deviation over $L \in \{8, 12, 16, 20, 24\}$ yields finite-size effective parameters $p_c^{\text{eff}} = 0.209 \pm 0.008$ and $\nu_{\text{eff}} = 1.72 \pm 0.10$ (bootstrap statistical uncertainty, 100 resamples). Both estimates are substantially above the thermodynamic values ($p_c \approx 0.16$, $\nu \approx 1.28$ [8]). These deviations are expected systematic effects of finite-size corrections; the quoted bootstrap uncertainties quantify statistical precision only and do not account for this systematic bias, which is the dominant source of deviation at $L \leq 24$. Reliable extraction of thermodynamic exponents requires system sizes up to $L \sim \mathcal{O}(100)$ – $\mathcal{O}(500)$ [8], far beyond what is accessible here. At these system sizes, the five-curve collapse qualitatively captures the correct scaling structure, demonstrating that the simulation reproduces the qualitative phenomenology of the MIPT universality class.

TABLE I. Finite-size effective critical quantities from simulation vs. literature. System sizes $L \in \{8, 12, 16, 20, 24\}$, 200 samples per point. Reported $\pm$ values are bootstrap statistical errors only; systematic finite-size bias is the dominant source of deviation from thermodynamic values for p_c^{eff} , ν_{eff} , and α_{eff} .

Quantity	Finite-size eff. ($L \leq 24$)	Literature
p_c^{eff} (curve crossing)	0.203 ± 0.017	0.16 [1]
p_c^{eff} (FSS collapse)	0.209 ± 0.008	0.16 [1]
ν_{eff}	1.722 ± 0.104	1.28 [8]
α_{eff} (at $p = 0.16$)	0.270 ± 0.074	≈ 1.6 [1]
$\alpha_{\text{vol}} (p = 0)$	0.995 ± 0.017	~ 1 (system dep.)
$S_{\infty} (L = 24, p = 0.5)$	0.105 ± 0.022	$\mathcal{O}(1)$

DISCUSSION

Quantum Error Correction Threshold Analogy

The MIPT is formally analogous to the quantum error correction (QEC) threshold theorem. In the area-law phase ($p > p_c$), measurements destroy quantum information faster than unitary scrambling can protect it: an adversary who monitors all measurement outcomes can, in principle, reconstruct the logical state from the syndrome. In the volume-law phase ($p < p_c$), scrambling delocalizes quantum information so thoroughly that even complete knowledge of measurement outcomes is insufficient to reconstruct the logical state [9]. The critical rate p_c is therefore closely related to the maximum error rate below which quantum information can in principle be preserved—a structural analog of the fault-tolerance threshold. This connection to QEC motivates practical interest in MIPT physics: understanding how p_c scales with circuit architecture has direct implications for the design of robust quantum error-correcting codes.

The Log-Scaling Exponent and the Replica Field Theory

At $p = p_c$, entanglement scales as $S(L/2) = \alpha(p_c) \ln(L/2) + \text{const}$ with $\alpha \approx 1.6$ [1] (established at $L \gg 100$). A standard unitary (1+1)D CFT with central charge c would predict $\alpha = c/6$ for a system with open boundaries (Calabrese and Cardy [10]), which would give $c \approx 9.6$ —an unphysical value for any conventional CFT.

The reason is that the MIPT critical point is *not* a standard unitary CFT. Jian *et al.* [7]—including course instructor Prof. Yi-Zhuang You as a co-author—map the MIPT to a two-dimensional replica statistical mechanics model in which the $n \rightarrow 1$ replica limit generates an effective non-unitary field theory. The coefficient α characterizes the scaling dimension of the twist operator at this non-unitary fixed point and has no direct inter-

pretation as a Virasoro central charge. Bao, Choi, and Altman [9] confirm this picture from a complementary random tensor network perspective. We therefore report α as a raw numerical exponent, explicitly avoiding the notation $c_{\text{eff}} = 6\alpha$ throughout this work.

Information Scrambling and the Correlation Length

Quantum scrambling is the process by which local information spreads under unitary evolution, becoming inaccessible to any local probe. In Clifford circuits, scrambling is quantified by the growth rate of the stabilizer generators' spatial support. Near the critical point, the correlation length diverges as $\xi \sim |p - p_c|^{-\nu}$, signaling a renormalization group fixed point. The competition between unitary scrambling and measurement-induced localization drives this divergence; sufficiently close to p_c , the system cannot distinguish the volume-law and area-law phases on a finite chain. Out-of-time-order correlators (OTOCs) [1] provide the standard diagnostic for this competition, measuring how much a local operator spreads under evolution; their behavior near p_c encodes the same universality class as the entropy. The thermodynamic exponent $\nu \approx 1.3$ [8] implies that the critical window in measurement rate narrows as $\Delta p \sim L^{-1/\nu}$ with increasing system size. Our finite-size effective value $\nu_{\text{eff}} = 1.72 \pm 0.10$ is still converging toward this thermodynamic value and should not be interpreted as evidence for a distinct universality class.

Structural Analogies with Threshold Phenomena in Machine Learning

The MIPT belongs to a broader class of *threshold phenomena in complex information-processing systems*: below a critical disturbance rate, structured information survives; above it, it irreversibly collapses. We note two speculative structural analogies, emphasizing that these are heuristic comparisons with no known quantitative mapping, not formal equivalences.

Grokking. In neural network training, “grokking” [11] refers to a sharp generalization transition—delayed memorization followed by sudden algorithmic compression—that emerges after prolonged training on small datasets. Both grokking and the MIPT exhibit a sharp crossover between two qualitatively distinct regimes as a single control parameter crosses a threshold; in both cases, the transition is accompanied by a structural reorganization of internal representations rather than a smooth interpolation. The analogy is qualitative only.

Robustness of learned representations under optimization pressure. Whether safety-relevant internal representations survive continued fine-tuning is structurally analogous to asking whether volume-law entanglement

survives increasing measurement rate. Near the critical point, the tuning window in p narrows as $\Delta p \sim L^{-1/\nu}$ as system size grows, suggesting that analogous robustness thresholds in ML systems may sharpen with model scale. We emphasize this comparison is heuristic: the MIPT exponent ν has no known quantitative relationship to learning dynamics. Information-delocalization measures—effective representation rank, participation ratio in the basis of internal representations—may nonetheless serve as diagnostic tools for studying representation robustness in mechanistic interpretability research, motivated by this analogy.

These analogies are speculative and are offered as motivation for future cross-disciplinary investigation, not as formal claims.

CONCLUSION

We implemented the Aaronson-Gottesman stabilizer tableau from scratch and used it to study measurement-induced phase transitions in one-dimensional random Clifford circuits. Simulating $L \in \{8, 12, 16, 20, 24\}$ with 200 disorder realizations per point, we reproduced the qualitative MIPT phenomenology: a crossover from volume-law ($S \propto L$) to area-law ($S = O(1)$) entanglement as the measurement rate increases past a finite-size effective critical rate $p_c^{\text{eff}} \approx 0.20$. All system sizes studied are too small for reliable thermodynamic exponent extraction. The extracted finite-size effective parameters ($p_c^{\text{eff}} \approx 0.20$, $\nu_{\text{eff}} \approx 1.72$, $\alpha_{\text{eff}} \approx 0.27$) are substantially affected by finite-size corrections and are not quantitatively consistent with thermodynamic literature values [1, 8]; the quoted bootstrap uncertainties are statistical only. The volume-law coefficient $\alpha_{\text{vol}} = 0.995 \pm 0.017$ and the area-law saturation $S_{\infty} = 0.105 \pm 0.022$ ebits are physically interpretable and independently verified by the physics audit (7/7 checks passed).

We discussed the formal analogy between p_c and the quantum error correction threshold, the replica-field-theory interpretation of the log-scaling exponent α (which is *not* a CFT central charge), and speculative structural analogies with grokking and representation robustness in machine learning.

Future directions include extending to two-qubit measurements and non-Clifford gates (requiring tensor network methods), implementing OTOC diagnostics to directly measure scrambling, studying the transition in open systems with non-Markovian noise, and making the analogy with ML threshold phenomena more precise via representation rank diagnostics applied to large language models.

The author thanks Prof. Yi-Zhuang You for lecture material motivating this investigation, and acknowledges Ref. [7] (co-authored by Prof. You) as a foundational analytical reference for the replica field theory of the

MIPT. Simulations used Claude Code (Anthropic) for AI-assisted implementation of the stabilizer tableau and circuit infrastructure.

Stabilizer Tableau Gate Update Rules

Following Aaronson and Gottesman [5] (Table 1), the gate update rules are applied simultaneously to all rows $h \in \{0, \dots, 2n - 1\}$ of the tableau. Let $x_{h,j}$ and $z_{h,j}$ denote the X - and Z -bits for qubit j in row h , and r_h the phase bit.

Hadamard on qubit a . The X - and Z -bits are swapped; the phase flips whenever both were 1:

$$\begin{aligned} r_h &\oplus= x_{h,a}^{\text{old}} \cdot z_{h,a}^{\text{old}}, \\ x_{h,a} &\leftarrow z_{h,a}^{\text{old}}, \quad z_{h,a} \leftarrow x_{h,a}^{\text{old}}. \end{aligned} \quad (5)$$

The copy of $x_{h,a}^{\text{old}}$ must be saved before any modification.

Phase (S) gate on qubit a . The phase flips when $x_{h,a} = 1$ and the old $z_{h,a} = 1$ (i.e., the generator has Y on qubit a); then $z_{h,a}$ is toggled:

$$\begin{aligned} r_h &\oplus= x_{h,a} \cdot z_{h,a}^{\text{old}}, \\ z_{h,a} &\oplus= x_{h,a}. \end{aligned} \quad (6)$$

One can verify: $S|Y\rangle \rightarrow -|X\rangle$, i.e., $Y \mapsto -X$, consistent with $SY S^\dagger = -X$.

CNOT with control a , target b .

$$\begin{aligned} r_h &\oplus= x_{h,a} \cdot z_{h,b} \cdot (x_{h,b} \oplus z_{h,a} \oplus 1), \\ x_{h,b} &\oplus= x_{h,a}, \\ z_{h,a} &\oplus= z_{h,b}. \end{aligned} \quad (7)$$

All three updates use copies of the pre-update columns; the order of the last two may be exchanged freely.

Rowsum phase function. The measurement algorithm requires computing the product of two Pauli operators. The phase contribution is [5]:

$$g(x_1, z_1, x_2, z_2) = \begin{cases} z_2 - x_2 & x_1 = 1, z_1 = 1, \\ z_2(2x_2 - 1) & x_1 = 1, z_1 = 0, \\ x_2(1 - 2z_2) & x_1 = 0, z_1 = 1, \\ 0 & x_1 = 0, z_1 = 0, \end{cases} \quad (8)$$

and the new phase bit of row h after adding row i is:

$$r_h \leftarrow \left\lfloor \frac{1}{2} \left(2r_h + 2r_i + \sum_j g_{ij} \right) \bmod 4 \right\rfloor, \quad (9)$$

where $g_{ij} \equiv g(x_{i,j}, z_{i,j}, x_{h,j}, z_{h,j})$ and the sum runs over $j = 0, \dots, n-1$. The sum runs over all n qubits and must be computed in ordinary (mod-4) integer arithmetic—not mod-2.

GF(2) Rank Algorithm

Entanglement entropy requires computing $\text{rank}_{\text{GF}(2)}(M_B)$, the rank of the binary matrix M_B over the field with two elements. **numpy.linalg.matrix_rank operates over $\mathbb{R}$ and cannot be used here:** floating-point arithmetic does not respect GF(2) arithmetic, and it returns incorrect ranks for binary matrices with linearly dependent rows that sum to zero modulo 2.

The correct algorithm is Gaussian elimination with XOR operations:

1. Copy M_B to a working matrix M of dtype `int8`.
2. Initialize `rank` = 0 and `pivot_row` = 0.
3. For each column $c = 0, 1, \dots, 2|B| - 1$:
 - (a) Find the first row $r \geq \text{pivot_row}$ with $M[r, c] = 1$. If none, skip to the next column.
 - (b) Swap rows r and `pivot_row`.
 - (c) For every other row $r' \neq \text{pivot_row}$ with $M[r', c] = 1$: set $M[r'] \oplus= M[\text{pivot_row}]$ (XOR, element-wise).
 - (d) Increment `rank` and `pivot_row`.
4. Return `rank`.

XOR ($\oplus$) implements addition in GF(2) without modular reduction overhead. The algorithm runs in $O(n \cdot |B|^2)$ time, which is negligible compared to the $O(n^2)$ cost of a single measurement operation.

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